

ENGR 213: Applied Ordinary Differential Equations

Master Study Guide & Curriculum-Grounded Textbook Companion

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Primary Textbook:

Advanced Engineering Mathematics

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Master Course Roadmap & Curriculum Matrix

Textbook Section	Lecture Topic & Scope	Governing Mathematical Engine	Key Exam Archetype
§1.1 & §1.2	Definitions, Terminology & IVPs	Order, Linearity, Normal Form $\frac{dy}{dx} = f(x, y)$, Picard Existence	Continuity of f and $\frac{\partial f}{\partial y}$ at (x_0, y_0)
§2.1	Solution Curves Without a Solution	Direction Fields, Autonomous DEs, Phase Lines	Classifying attractors, repellers & semi-stable nodes
§2.2	Separable Equations	$\int \frac{dy}{h(y)} = \int g(x)dx + C$	Tracking singular lost solutions when $h(y) = 0$
§2.3	First-Order Linear Equations	Integrating Factor $\mu(x) = e^{\int P(x)dx}$, $\frac{d}{dx}[\mu y] = \mu Q$	Standardizing leading coefficient to 1
§2.4	Exact Differential Equations	Test $M_y = N_x$, Potential function $F(x, y) = C$	Non-exact integrating factors $\mu(x)$ or $\mu(y)$
§2.5	Solutions by Substitutions	Homogeneous ($y = ux$), Bernoulli ($u = y^{1-n}$), Linear ($u = Ax + By + C$)	Bernoulli linearization to 1st-order linear
§2.7 & §2.8	Linear & Non-Linear Physical Models	Mass balance $\frac{dA}{dt} = R_{in} - R_{out}$, Logistic $\frac{dP}{dt} = rP(1 - P/K)$	Draining brine tanks with dynamic liquid volumes
§17.1 & §17.2	Complex Numbers & Powers	Polar form $z = re^{i\theta}$, De Moivre's $[re^{i\theta}]^n = r^n e^{in\theta}$	Finding all n -th roots on symmetric circles
§3.1 & §3.3	Constant-Coefficient Linear Equations	Characteristic Eq $ar^2 + br + c = 0$, Wronskian $W \neq 0$	3 Cases: Distinct real, repeated, complex conjugate
§3.4 & §3.5	Non-Homogeneous Equations	Undetermined Coefficients (Table/Annihilator), Variation of Params	Duplication rule (x^k) and universal Wronskian u_1, u_2
§3.6 & §3.7	Cauchy-Euler & Reduction of Order	Equidimensional $ax^2y'' + bxy' + cy = 0$, $y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx$	Auxiliary eq $am(m-1) + bm + c = 0$
§3.8	Mechanical & Electrical Oscillators	Mass-Spring $mx'' + \beta x' + kx = F(t)$, RLC Circuit	Underdamped envelope decay, resonance envelope $t \sin(\omega t)$
§4.1 – §4.3	The Laplace Transform Engine	$\mathcal{L}f(t) = \int_0^\infty e^{-st} f(t)dt$, Frequency shift $\mathcal{L}e^{at}f = F(s-a)$	Solving 2nd-order IVPs via algebraic partial fractions
§5.1.2	Power Series Solutions	$y = \sum c_n x^n$ about ordinary points, Recurrence relations	Aligning index shifts to x^k and peeling off terms
§10.1 – §10.4	Systems of Linear DEs	Normal form $\mathbf{X}' = \mathbf{A}\mathbf{X}$, Eigenvalues $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$	Phase plane trajectories, nodes, saddles, centers, spirals

The 5 Fundamental Analytical Solution Engines

1. The Separation & Substitution Engine

When equations are non-linear or non-separable, substitutions convert them into standard separable or linear forms:

- **Homogeneous Functions:** If $f(tx, ty) = f(x, y)$, substitute $y = ux \implies dy = udx + xdu$. Then:

$$u + x \frac{du}{dx} = f(1, u) \implies \frac{du}{f(1, u) - u} = \frac{dx}{x}$$

- **Bernoulli Equations:**

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \xrightarrow{w=y^{1-n}} \frac{dw}{dx} + (1-n)P(x)w = (1-n)Q(x)$$

- **Linear Composition:** $\frac{dy}{dx} = f(Ax + By + C) \xrightarrow{u=Ax+By+C} \frac{1}{B} \left(\frac{du}{dx} - A \right) = f(u) \implies \frac{du}{A+Bf(u)} = dx$

2. The Exact Potential Engine

For differential form $M(x, y)dx + N(x, y)dy = 0$:

1. Calculate $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$.
2. If $M_y = N_x$, exactness holds. Integrate M with respect to x :

$$F(x, y) = \int M(x, y)dx + g(y)$$

3. Differentiate with respect to y and set equal to $N(x, y)$:

$$\frac{\partial F}{\partial y} = \frac{\partial}{\partial y} \left[\int M dx \right] + g'(y) = N(x, y)$$

4. Solve for $g'(y)$, integrate with respect to y , and state implicit solution $F(x, y) = C$.
5. If non-exact, find integrating factor:

$$\mu(x) = \exp \left(\int \frac{M_y - N_x}{N} dx \right) \quad \text{or} \quad \mu(y) = \exp \left(\int \frac{N_x - M_y}{M} dy \right)$$

3. The Second-Order Constant-Coefficient Engine

For $ay'' + by' + cy = g(x)$:

1. **Solve Homogeneous Equation:** $ar^2 + br + c = 0$.
 - $r_1 \neq r_2$: $y_c = c_1 e^{r_1 x} + c_2 e^{r_2 x}$
 - $r_1 = r_2 = r$: $y_c = c_1 e^{rx} + c_2 x e^{rx}$
 - $r = \alpha \pm i\beta$: $y_c = e^{\alpha x} (c_1 \cos(\beta x) + c_2 \sin(\beta x))$
2. **Find Particular Solution** y_p :

- **Undetermined Coefficients:** Use standard table. Multiply by x^k if any term duplicates y_c .
- **Variation of Parameters:**

$$y_p = -y_1 \int \frac{y_2 g(x)}{aW} dx + y_2 \int \frac{y_1 g(x)}{aW} dx, \quad W = y_1 y_2' - y_1' y_2$$

3. **Assemble General Solution:** $y(x) = y_c(x) + y_p(x)$. Apply initial conditions **only** to the full general solution!

4. The Laplace Operational Transform Engine

1. Transform differential equation using $\mathcal{L}y' = sY(s) - y(0)$ and $\mathcal{L}y'' = s^2Y(s) - sy(0) - y'(0)$.
2. Solve algebraically for $Y(s)$:

$$Y(s) = \frac{P(s)}{Q(s)}$$

3. Complete squares in quadratic denominators or decompose via partial fractions.
4. Apply Inverse Laplace Transform using shifting theorems $\mathcal{L}^{-1}F(s - a) = e^{at}f(t)$.

5. The Linear Systems & Eigenvalue Engine

For $\mathbf{X}' = \mathbf{A}\mathbf{X}$:

1. Compute eigenvalues: $\det(\mathbf{A} - \lambda\mathbf{I}) = 0 \implies \lambda^2 - \text{Tr}(\mathbf{A})\lambda + \det(\mathbf{A}) = 0$.
 2. For each λ , solve $(\mathbf{A} - \lambda\mathbf{I})\mathbf{v} = \mathbf{0}$ for eigenvector $\mathbf{v}$.
 3. General solution: $\mathbf{X}(t) = \sum c_i \mathbf{v}_i e^{\lambda_i t}$.
 4. If complex $\lambda = \alpha \pm i\beta$, split $\mathbf{v}e^{\lambda t} = (\mathbf{a} + i\mathbf{b})e^{\alpha t}(\cos \beta t + i \sin \beta t)$ into real and imaginary vector solutions.
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consequently it may be possible to see at a glance certain qualitative aspects of the solutions—regions in the plane, for example, in which a solution exhibits an unusual behavior. A single solution curve that passes through a direction field must follow the flow pattern of the field; it is tangent to a lineal element when it intersects a point in the grid.

EXAMPLE 1 Direction Field

The direction field for the differential equation $dy/dx = 0.2xy$ shown in **FIGURE 2.1.2(a)** was obtained using computer software in which a 5×5 grid of points (mh, nh) , m and n integers, was defined by letting $-5 \leq m \leq 5$, $-5 \leq n \leq 5$ and $h = 1$. Notice in Figure 2.1.2(a) that at any point along the x -axis ($y = 0$) and the y -axis ($x = 0$) the slopes

Figure M.1: Direction field and solution curves — from Zill 7th Ed. Chapter 2 (Fig. 2.1.2).

use the fact that $\frac{1}{4}A_0 = A(5730)$ or $\frac{1}{4}A_0 = A_0e^{5730k}$. The last equation implies $5730k = \ln \frac{1}{4} = -\ln 2$ and so we get $k = -(\ln 2)/5730 = -0.00012097$. Therefore $A(t) = A_0e^{-0.00012097t}$. With $A(t) = 0.001A_0$ we have $0.001A_0 = A_0e^{-0.00012097t}$ and $-0.00012097t = \ln(0.001) = -\ln 1000$. Thus

$$t = \frac{\ln 2}{0.00002867} \approx 24,180 \text{ years.} \quad \equiv$$

The date found in Example 3 is really at the border of accuracy of this method. The usual carbon-14 technique is limited to about 10 half-lives of the isotope, or roughly 60,000 years. One fundamental reason for this limitation is the relatively short half-life of C-14. There are other problems as well; the chemical analysis needed to obtain an accurate measurement of the remaining C-14 becomes somewhat formidable around the point 0.001A₀.

Figure M.2: Liquid mass balance in tank with dynamic volume — from Zill 7th Ed. Chapter 2 (Fig. 2.7.4).

$$x(t) = \frac{\sqrt{2}}{6} \cos(8t - (-0.245)) \quad \text{or} \quad x(t) = \frac{\sqrt{2}}{6} \cos(8t + 0.245). \quad (9')$$

■ Graphical Interpretation

FIGURE 3.8.4(a) illustrates the mass in Example 2 going through approximately two complete cycles of motion. Reading left to right, the first five positions marked with black dots in the figure correspond, respectively,

- to the initial position ($x = \frac{2}{3}$) of the mass below the equilibrium position,
- to the mass passing through the equilibrium position ($x = 0$) for the first time heading upward,

Figure M.3: Damped mechanical oscillator decay curves (Underdamped vs Critically Damped vs Overdamped) — from Zill 7th Ed. Chapter 3 (Fig. 3.8.4).

corresponding entries in the vector of the left, we obtain the more familiar statement

$$x = c_1 e^{-t} + 3c_2 e^{4t}, \quad y = -c_1 e^{-t} + 2c_2 e^{4t}.$$

As pointed out in Section 10.1, we can interpret these equations as parametric equations of a curve or **trajectory** in the xy -plane or **phase plane**. The three graphs shown in **FIGURE 10.2.1**, $x(t)$ in the tx -plane, $y(t)$ in the ty -plane, and the trajectory in the phase plane, correspond to the choice of constants $c_1 = c_2 = 1$ in the solution. A collection of trajectories in the phase plane as shown in **FIGURE 10.2.2** is said to be a **phase portrait** of the given linear system. What appears to be two red lines in Figure 10.2.2 are

Figure M.4: Trajectories in phase plane for 2x2 linear system — from Zill 7th Ed. Chapter 10 (Fig. 10.2.2).

⚡ Master Formula & Theorem Reference Card

Mathematical Law	Formula / Criterion	Essential Exam Notes
Separable ODE	$\int \frac{1}{h(y)} dy = \int g(x) dx + C$	Check $h(y) = 0$ for singular solutions
Integrating Factor (1st-Order)	$\mu(x) = \exp\left(\int P(x) dx\right)$	Ensure leading coefficient of y' is 1
Exactness Criterion	$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$	Solution is implicit potential $F(x, y) = C$
Non-Exact Multipliers	$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}, \quad \mu(y) = e^{\int \frac{N_x - M_y}{M} dy}$	Expression must be purely single-variable
Bernoulli Linearization	$w = y^{1-n} \implies w' + (1-n)Pw = (1-n)Q$	Divide through by y^n first
Wronskian Determinant	$W(y_1, y_2) = y_1 y_2' - y_1' y_2$	$W \neq 0 \iff$ Linearly independent
Cauchy-Euler Auxiliary Eq	$am(m-1) + bm + c = 0$	Solution bases are $x^m, x^m \ln x, x^\alpha \cos(\beta \ln x)$
Mass-Spring Quasi-Frequency	$\omega_d = \sqrt{\frac{k}{m} - \left(\frac{\beta}{2m}\right)^2}$	Exists only when underdamped ($\beta^2 < 4mk$)
Pure Resonance Growth	$x(t) = \frac{F_0}{2m\omega_0} t \sin(\omega_0 t)$	Amplitude grows linearly with time
Laplace 1st Shift Theorem	$\mathcal{L}e^{at} f(t) = F(s-a)$	Complete the square in denominator
Laplace Derivative Formulas	$\mathcal{L}y'' = s^2 Y - sy(0) - y'(0)$	Embeds initial conditions algebraically
De Moivre's Theorem	$[r(\cos \theta + i \sin \theta)]^n = r^n(\cos n\theta + i \sin n\theta)$	Roots spaced by $\Delta\theta = \frac{2\pi}{n}$

🚫 The 6 Fatal Exam Traps in ENGR 213

- The Premature Initial Condition Error:** Never plug $y(x_0) = y_0$ into the complementary solution $y_c(x)$ before finding the particular solution $y_p(x)$. Initial conditions apply **only to the full general solution** $y(x) = y_c(x) + y_p(x)$!
- The Non-Standard Integrating Factor Trap:** In $xy' + 2y = 4x^2$, $P(x)$ is **not** 2; it is $\frac{2}{x}$! You must divide by x before calculating $\mu(x)$.
- The Tank Volume Blindspot:** If $r_{\text{in}} \neq r_{\text{out}}$, volume is dynamic: $V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t$. Treating volume as constant destroys the differential equation.
- The Annihilator Duplication Trap:** If $g(x)$ matches any term in y_c , you must multiply your trial particular solution by x (or x^2). If you do not, the left-hand side will annihilate to 0, producing a nonsense equation like $0 = 4e^{2x}$.
- The Missing Generalized Eigenvector:** In a repeated eigenvalue system where only one independent eigenvector $\mathbf{v}$ exists, the second solution is $(\mathbf{v}t + \mathbf{u})e^{\lambda t}$, where $(\mathbf{A} - \lambda\mathbf{I})\mathbf{u} = \mathbf{v}$. Writing $\mathbf{v}te^{\lambda t}$ is fatally

incorrect.

6. **The Inverse Laplace Partial Fraction Sign Trap:** In $\frac{s}{(s+2)^2+9}$, split the numerator into $(s+2) - 2$. Then $\mathcal{L}^{-1} = e^{-2t} \cos(3t) - \frac{2}{3}e^{-2t} \sin(3t)$. Watch the sign of the exponential shift!